

Data Warehousing and Modeling

1 What is a Data Warehouse?

W. H. Inmon's Definition

"A data warehouse is a **subject-oriented**, **integrated**, **time-variant**, and **nonvolatile** collection of data in support of management's decision-making process."

It serves as a decision support database maintained separately from the organization's operational databases, providing a solid platform of consolidated, historical data for analysis.

1.1 The Four Characteristics

1. Subject-Oriented:

- Organized around major subjects, such as **customer**, **product**, **sales**.
- Focuses on modeling and analysis for decision makers, not on daily operations or transaction processing.
- Provides a simple and concise view by excluding data that are not useful in the decision support process.

2. Integrated:

- Constructed by integrating multiple, heterogeneous data sources (relational databases, flat files, on-line transaction records).
- Data cleaning and integration techniques are applied to ensure consistency in naming conventions, encoding structures, and attribute measures.
- *Example:* Reconciling hotel prices that might have different currency, tax, or breakfast coverage across sources.

3. Time-Variant:

- The time horizon is significantly longer than operational systems (e.g., **5–10 years** vs. current value data).
- Every key structure contains an element of time, explicitly or implicitly.
- Operational data may or may not contain a "time element" in its key, but warehouse data always does to provide a historical perspective.

4. Non-Volatile:

- A physically separate store of data transformed from the operational environment.

- Operational updates do not occur here; it does not require transaction processing, recovery, or concurrency control.
- Requires only two operations in data accessing: **initial loading** and **access**.

2 Data Warehouse vs. Other Systems

2.1 DW vs. Heterogeneous DBMS

- **Traditional approach (Query-driven):** Uses wrappers/mediators. Queries are translated on-the-fly. It is complex and competes for local resources.
- **Data Warehouse (Update-driven):** Information is integrated in advance and stored for direct high-performance query and analysis.

2.2 OLTP vs. OLAP

Data Warehouses perform **OLAP** (Analytical), while traditional RDBMS perform **OLTP** (Transactional).

Feature	OLTP	OLAP
User	Clerk, IT professional	Knowledge worker (manager, analyst)
Function	Day-to-day operations	Decision support
DB Design	Application-oriented (E-R)	Subject-oriented (Star, Snowflake)
Data	Current, isolated, detailed	Historical, integrated, consolidated
Access	Read/write, index/hash on PK	Mostly read, complex queries

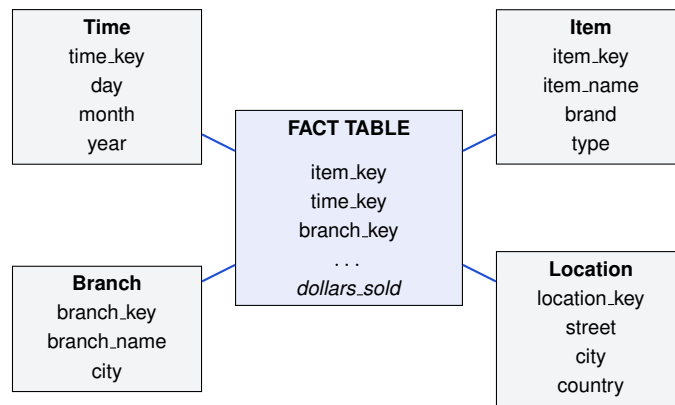
2.3 Why Separate them?

- **Performance:** Operational DBs are tuned for fast updates/concurrency; Warehouses are tuned for complex multidimensional queries.
- **Data Quality:** Consolidating data from different sources requires reconciliation of inconsistent representations and formats.

3 Conceptual Modeling: Schemas

3.1 Star Schema

A single **Fact Table** in the middle is connected to a set of **Dimension Tables**. It is simple and highly efficient for queries.



3.2 Snowflake or Fact Constellation

- **Snowflake Schema:** A refinement of Star Schema where dimension tables are normalized, splitting data into additional tables (forming a snowflake shape).
- **Fact Constellation (Galaxy):** Multiple fact tables share dimension tables. This is a collection of stars.

Data Warehousing transforms raw operational data into a strategic business asset.